Lucie Correia - Seeking a 3-months internships starting in June.

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Education

Telecom Nancy - Engineering School in Nancy, France **2023 - 2026**
MSc in Computer Science and Cybersecurity - 3,7/4 GPA

Key coursework : Operational research, Algorithms, Artificial Intelligence (Markov chains, Machine Learning, Deep Learning), Finance (Top Student), Software Security, Operating Systems, Programming, Networks.

UPC - Polytechnic University of Catalonia, Spain **2025 - 5 months**

Erasmus Student **:** Convex Optimization, Machine learning, Pentest, Data Privacy with R, Business & Finance.

Buffon High School Paris, France - CPGE MPSI/MP* - Preparatory classes **2021 - 2023**

Intense two-year preparatory course in Mathematics and Physics for entrance to French Grandes Ecole.

Experiences

Founder of a YouTube channel (ITeaser) **Since 2023**

Founded and animated a YouTube channel focused on interviews about computer science projects. Managed video editing and communication strategies, surpassing 20k views in just a few months.

Data Analyst Intern, GS1 France - National Organization for Barcodes **Summer 2024**

Contributing to the Digital Product Passport (DPP) project by designing Excel solutions and producing detailed reports.

Secretary, Cybersecurity Club (Hacking'TN) **2024 - 1 year**

Created medium-level CTF challenges (programming, networks, web, osint) for a self-hosted competition and participated in CTF competitions such as GreHack, TRACS and LeHack Paris, achieving top 6 rankings with my team.

Vice Treasurer, Junior Enterprise (TNS) **2024 - 1 year**

Managed a 10k budget, developed forecast budgets, supported commercial development, and contributed to strategy.

IT Projects

HFT - 2025 : Developing a C++/Python high-frequency trading system using WebSockets (Binance API) and NATS for messaging. Performance monitoring via Prometheus/Grafana with backtesting and risk analysis (VaR, Sharpe Ratio).

Attacks Detector - 2025 : Developed an intrusion detection system for DDoS using ML (SVM, ANN, Decision Trees) with Keras/TensorFlow. Processed data via InfluxDB/Telegraf. Deployed the system with Docker on GCP (Cloud Run).

Trading algorithm - 2025 : Developed and backtested a Python trading algorithm and a fine-tuned model to predict Bitcoin prices with the IBM TTM model. *Time Series, Zero-Shot, Simple Moving Averages, RMSE, large dataset.*

Web App - 2025 : Designed a Python application for option pricing with automated reporting. *Black-Scholes, Monte Carlo.*

Home Server - 2024 : Configured and managed a personal NAS with Unraid OS, hosting services. *Networks.*

Compiler - Group Project in 2024 : Developed a Mini-Python Compiler in Java with LL1 grammar. *Lexer & Parser.*

Skills

Programming Language: Python (NumPy, Flask, Pandas, CVXPY, Scapy, pwntools, TensorFlow, Scikit), Java, C, C++.

Quantitative Finance: Stochastic Calculus, Time Series Analysis, Financial Models, TradingView, Bloomberg Terminal.

Software: Git, Docker, Bash, GCP (Cloud Run), VirtualBox, Linux, Wireshark, Nmap, Metasploit, Burp Suite, Postman.

Databases & Reporting : SQL, SQLite, Excel, VBA, Power BI, Latex, Jupyter Notebook.

Soft Skills: Leadership, Entrepreneurship, Analytical Thinking, Problem-Solving, Teamwork, Communication.

Certifications

Introduction to Trading, Machine Learning & GCP : Machine Learning, GCP, Vertex AI, Big Query, Finance.

Quantitative Research Job Simulation by JPMorgan : Market Forecasting of Gas Market in Python, Credit Risk Analysis.

Advanced Portfolio Construction by EDHEC **:** Factorial Investment, Risks and Asset optimization in Python.

Financial Markets by Yale University **:** Investment Management, Portfolio Theory, stocks & bonds, CAPM, Fama-French.

Docker Essentials by IBM **:** Dockerfile, Docker Hub, CI/CD.

TOEIC - English Proficiency : 890/990 (B2 Level) for listening and reading.

Project Gestion by Centrale Lille **:** Agile Methodology.